

From Noisy Conflict Scores to Forecastable Risk States in Social Media Comment Streams

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Abstract—Social media conflict forecasting promises earlier intervention than post-hoc content classification, yet raw text-derived scores are sparse and temporally noisy. We construct a conflict index without task-specific labels from three interpretable components—attack intensity from a multilingual toxicity detector, negative-arousal emotion from a zero-shot NLI anger classifier, and stance polarization from semantic camp clustering—and introduce a leakage-resistant cleaning pipeline that regularizes the 12-hour time axis, learns all calibration and cleaning statistics from training data only, and performs causal exponential smoothing. A logistic model predicts whether the stabilized state will exceed the topic-specific training-period 80th percentile within the next six bins. Across 20 Zhihu topics and 1,706 test windows, it achieves AUC 0.897 and F1 0.767 against a persistence baseline of F1 0.764, with a significant recall gain (+0.098; 95% CI [0.042, 0.167]) and stable performance on entirely unseen topics (5-fold topic-held-out F1 0.787 ± 0.020 vs. persistence 0.705). The attack component is validated on 191,634 human Wikipedia Detox labels (AUC 0.988, Spearman $\rho = 0.507$); leave-one-out ablation identifies it as the most informative component. The same pipeline transfers to 20 Weibo reversal events (AUC 0.989), and a reproduction of pooled random splits quantifies R^2 inflation from protocol leakage on real data.

Index Terms—conflict-index construction, data cleaning, causal smoothing, risk-state prediction, social media comments, temporal split

I. INTRODUCTION

A. Challenges in Governing Social Media Public Opinion

The ubiquity of social media has transformed how information is produced, circulated, and contested. Public debates that once unfolded over days now escalate within hours as platforms enable rapid information cascades and viral diffusion [1]. Seemingly marginal topics can ignite large-scale opinion swings with tangible offline consequences. For example, coordinated online discourse has been shown to correlate with mass protests and even violent incidents in subsequent days [2], [3].

1) *Information Cascades and Offline Spillovers*: Information spreads faster than it can be verified. Emotional contagion and algorithmic amplification jointly fuel cascades that can spill over into offline actions. Once critical mass is reached, moderation or fact-checking becomes less effective, as narratives solidify through repeated exposure [1]. These cascades thus demand proactive, not reactive, management.

2) *Delays in Post-hoc Sentiment Analysis*: Conventional sentiment analysis frameworks primarily offer retrospective insight: they describe how people felt after events unfolded. However, by the time significant polarity shifts are detected, the discussion may already have escalated beyond control. This latency between event occurrence and analytical response undermines the capacity for early intervention [4].

3) *Research Gaps in Early Warning*: Despite advances in opinion mining and social listening, few studies explicitly target early warning. Existing tools classify emotions or toxicity per message, but rarely attempt to forecast when online discontent is likely to escalate [4]. This absence of anticipatory modeling creates a crucial gap between detection and prevention. At the same time, converting a post-level score into a forecast target introduces an often overlooked assumption: the aggregated semantic signal must exhibit stable temporal dependence. A forecasting model cannot recover predictive structure that is absent from its input trajectory.

B. Limitations of Existing Technical Paradigms

1) *Lexicon/Rule-Based Systems: Cross-Language and Sarcasm Fragility*: Lexicon-based or rule-driven systems offer interpretability but struggle with context-dependent semantics, sarcasm, and multilingual variation. Irony (“Great, another perfect policy!”) or cultural idioms can easily invert meaning [5]. Moreover, such systems require language-specific dictionaries that fail in cross-lingual or code-mixed environments [6].

2) *Static Classifiers: Concept Drift and Out-of-Distribution Failure*: Traditional supervised classifiers, like SVM or CNN, are trained on fixed datasets and assume distributional stability. In dynamic online environments, vocabulary, sentiment norms, and social topics evolve rapidly—an issue known as concept drift [7]. Without continuous adaptation, static models degrade, misclassify emergent patterns, or fail to generalize to unseen contexts.

3) *Post-Level Semantic Analysis without Temporal Escalation Modeling*: Recent pretrained language models and text analysis systems substantially improve post-level semantic understanding, sentiment recognition, and harmful-content detection. However, these methods are often used to score or classify comments independently, without explicitly modeling how conflict accumulates and evolves across time. Consequently, they provide informative local signals but limited support for topic-level escalation forecasting and early warning. This motivates the introduction of a temporal forecasting layer on top of interpretable post-level risk signals [8].

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C. Contributions of This Work

To examine this assumption, we construct a conflict index without task-specific labels and evaluate when its resulting trajectories can and cannot be forecast by supervised temporal models. Rather than presenting a new neural architecture as the central contribution, we focus on signal stability, temporal dependence, and leakage-resistant evaluation. The main contributions are as follows:

(I-C1) Conflict-index quantification framework (Sec. III-B): We operationalize a candidate *conflict index* by combining attack-related negativity, negative high-arousal emotion, and stance polarization into a unified post-level score without task-specific conflict labels, using a genuine multilingual toxicity detector for attack intensity and a zero-shot NLI anger classifier for emotion. This unsupervised construction stage is distinct from the downstream forecasters, which are trained with historical trajectory targets. Its attack component is validated against 191,634 human-annotated English Wikipedia Detox comments (Spearman $\rho = 0.507$, AUC 0.988), the two stance camps are stable and balanced across topics (mean adjusted Rand index 0.95), and leave-one-out ablation identifies the attack component as the most informative.

(I-C2) Causal signal cleaning and risk-state formulation (Sec. IV-C): We regularize sparse topic streams on their true time axis, learn all clipping, imputation, scaling, and risk thresholds from training data only, and use causal smoothing to separate persistent risk states from bin-level semantic noise. Validation data select the smoothing scale and the smallest feature set within 0.005 F1 of the best candidate, without accessing the final test period.

(I-C3) Positive real-data risk-state prediction with boundary analysis (Sec. IV-M): On held-out Zhihu periods, the selected classifier achieves AUC 0.897 with a significant recall gain over persistence, and—unlike exact-value regression—generalizes to entirely unseen topics (5-fold topic-held-out F1 0.787 ± 0.020 vs. persistence 0.705). Five strong baselines (BiLSTM, GRU, TCN, Transformer, XGBoost) under the identical protocol fail to beat the simple logistic model, showing that signal quality, not architecture, determines performance. The pipeline also transfers to 20 Weibo reversal events (AUC 0.989). Threshold and feature ablations show that causal state history drives the gain, while exact-value regression remains persistence-dominated.

D. Paper Structure Overview

The remainder of this paper is organized as follows. Section II reviews related studies in sentiment analysis, harmful content detection, and temporal modeling. Section III presents conflict-index construction and the two temporal prediction tasks, including the leakage-free stance camps and ECDF calibration. Section IV reports the experiments: synthetic continuous forecasting, real-data risk-state prediction with signal cleaning, cross-platform transfer, measurement validation, ablation studies, boundary analysis, and limitations.

II. RELATED WORK

A. Deep Learning for Sentiment and Emotion Understanding

Deep learning has significantly advanced sentiment analysis by learning contextual and compositional linguistic patterns that are difficult to encode with lexicon-based rules or hand-crafted features. Recent surveys [9], [10] trace this evolution from classical approaches to neural architectures, highlighting how LSTMs, CNNs, and Transformer encoders (e.g., BERT and GPT-style models) improve classification accuracy through representation learning. Despite these gains, key challenges remain, including robustness to sarcasm and irony, domain shift across platforms and topics, multilingual and code-mixed text processing, and the need for explainable sentiment reasoning for high-stakes governance applications [9]. Recent work also moves beyond categorical sentiment labels toward continuous intensity scoring—for instance, SafeSpeech [11] uses integrated gradients to assign fine-grained hate-intensity scores to individual words, enabling more nuanced risk assessment than binary classification. These limitations motivate models that (i) provide interpretable intermediate signals and (ii) remain effective under rapidly changing discourse.

B. Risk Scoring for Harmful Content and Hazardous Conversations

Beyond categorical sentiment labels, risk scoring aims to assign continuous severity estimates to online comments to support scalable moderation and hazard monitoring. Wulczyn *et al.* [12] construct a large corpus of Wikipedia discussion comments annotated for personal attacks and train predictive models that output, for each comment, the probability of being an attack. Such probabilistic outputs serve as practical risk scores for triage and prioritization, enabling moderators to focus on high-risk items and to identify discussion spaces with elevated aggression. However, most risk-scoring systems remain *instance-centric*: they optimize snapshot detection at the comment level and treat posts as independent samples. As a result, they do not explicitly characterize how risk accumulates and propagates within a topic over time, nor do they naturally support trajectory-level forecasting and early warning. The conflict-index formulation in this paper builds on the idea of continuous risk scoring [11] while explicitly designing the score to be both post-level interpretable and suitable as a temporal signal for escalation modeling.

C. Temporal Forecasting of Public Opinion Trends with LSTM Variants

Forecasting public opinion dynamics during emergencies requires temporal models that can capture persistence, volatility, and bursty surges in online activity. LSTM-based predictors are widely adopted for such sequence modeling, but their performance can be sensitive to hyperparameter choices under highly nonlinear and nonstationary trends. To improve forecasting stability, Mu *et al.* [13] propose an Improved Particle Swarm Optimization LSTM (IPSO-LSTM) framework that uses adaptive search strategies (e.g., nonlinearly decreasing inertia weights and mutation mechanisms) to tune LSTM hyperparameters and

avoid poor local optima. Their results show improved error metrics (e.g., MAPE and RMSE) relative to standard LSTM variants on multiple emergency-event datasets [13]. While these studies demonstrate the value of enhanced temporal predictors, they typically operate on aggregate time series and do not incorporate fine-grained, post-level semantic evidence. Our work complements this line of research by coupling contextual language understanding (via sentence-embedding-derived conflict signals) with temporal sequence modeling (via LSTM) to produce interpretable, topic-level escalation trajectories and short-horizon risk forecasts.

Recent hybrid architectures further validate the CNN-LSTM paradigm for social-media event forecasting. Singh and Jain [14] combine DistilBERT embeddings with CNN-LSTM layers to predict civil unrest events from Twitter data, demonstrating cross-region generalizability. Upadhyay *et al.* [15] propose a self-attention BiLSTM with multi-scale CNN for crisis event detection, achieving strong F1 scores across benchmark datasets. On the theoretical side, Liu *et al.* [16] identify multi-scale memory cycles in public opinion time series, providing empirical support for the sliding-window formulation with carefully chosen lookback length L and forecast horizon H . For crisis-specific forecasting, Lou *et al.* [17] demonstrate that CNN-LSTM hybrid architectures outperform standalone CNN, LSTM, and Transformer variants by 4–6% on social-network crisis prediction tasks. We also note the emergence of efficient Transformer architectures for long-sequence forecasting, such as Informer [18], which uses ProbSparse self-attention to achieve $O(L \log L)$ complexity; these methods are primarily designed for sequences with hundreds to thousands of time steps, whereas our setting involves relatively short lookback windows ($L=12$) where lightweight recurrent architectures remain competitive.

D. Information Cascade and Diffusion Prediction

Beyond post-level content analysis, modeling how information spreads through social networks is essential for understanding conflict escalation dynamics. Recent graph-based approaches have advanced cascade prediction: Xu *et al.* [19] propose D², a two-stage GNN framework that first predicts future propagation paths from limited early-stage data, then detects rumors via neural architecture search over the completed graph. The same group extends this to hypergraph neural networks, in HyperIDP [20], jointly handling macroscopic cascade scale prediction and microscopic next-user forecasting with a cooperative-adversarial multi-task loss [?]. These works highlight the value of modeling diffusion structure for early detection; our work complements them by focusing on *content-driven* conflict signals rather than propagation topology, making the two approaches orthogonal and potentially synergistic.

III. CONFLICT-INDEX CONSTRUCTION AND TEMPORAL PREDICTION

A. Problem Definition

We study social-media comment streams organized by topic and discretized into fixed-length time bins. For a given topic, let the observation period be partitioned into T consecutive bins of equal duration. For notation simplicity, we omit the topic

index in the following definitions. Denote by $\mathcal{B}_t = \{x_{t,i}\}_{i=1}^{n_t}$ the multiset of n_t comments posted in bin $t \in \{1, \dots, T\}$, where each $x_{t,i}$ represents a text message.

Our first objective is to assign an interpretable *conflict index* to each comment via a text encoder f_θ :

$$c_{t,i} = f_\theta(x_{t,i}) \in [0, 1], \quad (1)$$

where higher values indicate stronger antagonistic cues (e.g., high-arousal negative emotion and confrontation markers). To obtain a topic-level risk signal per bin, we aggregate comment-level indices using a fixed aggregation operator $\text{Agg}(\cdot)$:

$$\bar{c}_t = \text{Agg}(\{c_{t,i}\}_{i=1}^{n_t}) \in [0, 1]. \quad (2)$$

Our second objective is to test whether this proxy is forecastable through short-horizon prediction. Given the historical binned sequence $\bar{c}_{1:t}$, we learn a temporal predictor g_ϕ to estimate the future trajectory $\hat{\mathbf{c}}$ over a horizon H :

$$\hat{\mathbf{c}}_{t+1:t+H} = g_\phi(\bar{c}_{1:t}). \quad (3)$$

An escalation warning is triggered at time bin t if the predicted risk is expected to exceed a predefined threshold η within the horizon, i.e.,

$$\max_{1 \leq h \leq H} \hat{c}_{t+h} \geq \eta, \quad (4)$$

optionally combined with a rapid-growth criterion to capture fast-rising trends.

B. Conflict-Index Quantification without Manual Labels

Our dataset contains only time-stamped comment text streams and provides no human-annotated conflict labels. We therefore construct a candidate unsupervised *conflict index* by combining three interpretable proxies: (i) attack-related negativity, (ii) negative high-arousal emotion intensity, and (iii) stance polarization derived from within-bin semantic bimodality. Because these components are not validated against task-specific annotations, the resulting quantity is an operational proxy whose construct validity must be evaluated separately from its temporal predictability. For each comment $x_{t,i}$ posted in time bin t , we compute

$$c_{t,i} = \sigma(w_a \tilde{a}_{t,i} + w_e \tilde{e}_{t,i} + w_s \tilde{s}_{t,i} + b) \in [0, 1], \quad (5)$$

where $\sigma(\cdot)$ is the sigmoid function, $w_a, w_e, w_s \geq 0$ control the relative contributions (e.g., $w_a : w_e : w_s = 0.5 : 0.3 : 0.2$), and b is a bias term (we use $b = 0$ in practice). The tildes denote *distribution-calibrated* scores (Sec. III-B4) so that components from different sources become comparable.

1) *Attack/Antagonism Intensity from a Pretrained Harm Model*: We obtain an antagonism score from a pretrained harmful/attack detector (e.g., toxicity/insult/hate model) that outputs a probability of attack-like content for a given text:

$$a_{t,i} = h_\psi(x_{t,i}) \in [0, 1], \quad (6)$$

where h_ψ is a frozen pretrained model. This score serves as a *pseudo-label prior* capturing confrontation markers, threats, harassment, and abusive language without requiring any task-specific annotation.

2) Emotion Intensity from a Zero-Shot Emotion Model:

We characterize emotionally charged negativity using a zero-shot (or pretrained) emotion model that produces probabilities for emotion categories. Let $p_{\text{neg}}(x_{t,i})$ denote the probability of negative sentiment and $p_{\text{anger}}(x_{t,i})$ denote the probability of anger (a high-arousal negative emotion). We define

$$e_{t,i} = \alpha p_{\text{neg}}(x_{t,i}) + (1 - \alpha) p_{\text{anger}}(x_{t,i}), \quad \alpha \in [0, 1], \quad (7)$$

where α balances general negativity and arousal-related anger (we use $\alpha \in [0.6, 0.8]$ in practice). This construction rewards comments that are both negative and emotionally intense, which are common precursors to conflict escalation. Recent empirical evidence from Weibo conflict discourse networks confirms that emotional similarity (affective homophily)—rather than structural prestige—is the dominant organizing principle in online conflict, validating the central role of negative high-arousal emotion in our conflict-index formulation [21].

3) Stance Polarization via Training-Period Semantic Camps:

In the absence of stance labels (pro/con/neutral), we estimate polarization by measuring whether the comments of a topic form two semantically separated camps. To keep the procedure leakage-free, we fit $k=2$ clustering (k-means) on the sentence embeddings of the training-period comments only and keep the two centroids fixed when scoring validation and test comments, mirroring the training-only fitting of the ECDF calibrators (Sec. III-B4). This per-topic formulation—rather than within-bin clustering—is required because many 12-hour bins contain too few comments for stable within-bin clustering. We first compute sentence embeddings using a frozen encoder $\text{Enc}(\cdot)$ (e.g., multilingual BERT pooling or sentence-transformer):

$$\mathbf{v}_{t,i} = \text{Enc}(x_{t,i}) \in \mathbb{R}^d. \quad (8)$$

The two training-period centroids $\mathbf{u}_1$ and $\mathbf{u}_2$ define the topic-level inter-camp separation

$$\Delta = 1 - \cos(\mathbf{u}_1, \mathbf{u}_2) \in [0, 2], \quad (9)$$

where $\cos(\cdot, \cdot)$ is cosine similarity. Next, we quantify how strongly each comment aligns with one camp over the other. Let $d(\cdot, \cdot)$ be a distance function (we use cosine distance). Define the camp-alignment confidence

$$q_{t,i} = \frac{|d(\mathbf{v}_{t,i}, \mathbf{u}_1) - d(\mathbf{v}_{t,i}, \mathbf{u}_2)|}{d(\mathbf{v}_{t,i}, \mathbf{u}_1) + d(\mathbf{v}_{t,i}, \mathbf{u}_2) + \epsilon} \in [0, 1], \quad (10)$$

with a small $\epsilon > 0$ to avoid division by zero. Finally, the stance polarization score for a comment is

$$s_{t,i} = \text{norm}(\Delta) \cdot q_{t,i} \in [0, 1], \quad (11)$$

where $\text{norm}(\Delta)$ maps the separation magnitude to $[0, 1]$ (e.g., $\text{norm}(\Delta) = \min(1, \Delta)$). This assigns higher polarization to topics that exhibit two well-separated training-period camps and to comments that are clearly affiliated with one camp. This unsupervised embedding-based approach to polarization measurement aligns with recent work on transformer-based echo chamber quantification [22] and unsupervised polarization knowledge extraction from text corpora [23].

Algorithm 1 Unsupervised Conflict Index Computation

Require: Comments $\{x_{t,i}\}$ grouped into bins $\mathcal{B}_t$; pretrained harm model h_ψ ; zero-shot emotion model g_ω ; embedding encoder Enc ; top fraction ρ

Ensure: Comment-level indices $\{c_{t,i}\}$ and bin-level trajectory $\{\bar{c}_t\}$

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1: for  $t = 1$  to  $T$  do
2:   for each comment  $x_{t,i} \in \mathcal{B}_t$  do
3:      $a_{t,i} \leftarrow h_\psi(x_{t,i})$  (attack prior)
4:      $(p_{\text{neg}}, p_{\text{anger}}) \leftarrow g_\omega(x_{t,i})$ 
5:      $e_{t,i} \leftarrow \alpha p_{\text{neg}} + (1 - \alpha) p_{\text{anger}}$ 
6:      $\mathbf{v}_{t,i} \leftarrow \text{Enc}(x_{t,i})$ 
7:   end for
8:   Cluster  $\{\mathbf{v}_{t,i}\}$  into  $k=2$  camps; get centroids  $\mathbf{u}_{t,1}, \mathbf{u}_{t,2}$ 
9:   Compute  $\Delta_t$  by (9)
10:  for each comment  $x_{t,i} \in \mathcal{B}_t$  do
11:    Compute  $q_{t,i}$  by (10); set  $s_{t,i} \leftarrow \text{norm}(\Delta_t) \cdot q_{t,i}$ 
12:  end for
13: end for
14: Calibrate  $\tilde{a}, \tilde{e}, \tilde{s}$  using ECDF mapping (12)
15: for all  $(t, i)$  do
16:   Compute  $c_{t,i}$  by fusion (5)
17: end for
18: for  $t = 1$  to  $T$  do
19:   Aggregate  $\bar{c}_t$  by top- $k$  mean (13)
20: end for

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4) *Distribution Calibration by Percentile Mapping:* Scores produced by different pretrained models and heuristics can have inconsistent scales and distributions. We therefore calibrate each raw component score using an empirical cumulative distribution function (ECDF) fitted only on comments in the temporal training segment of each topic. For a raw score $r_{t,i} \in [0, 1]$ (standing for $a_{t,i}$, $e_{t,i}$, or $s_{t,i}$), we define

$$\tilde{r}_{t,i} = \text{ECDF}_r(r_{t,i}) \in [0, 1], \quad (12)$$

which maps $r_{t,i}$ to its percentile rank. This calibration yields comparable $\tilde{a}_{t,i}, \tilde{e}_{t,i}, \tilde{s}_{t,i}$ for fusion in (5).

5) *Bin-Level Aggregation for Topic Risk Trajectories:* To obtain a topic-level risk signal per time bin, we aggregate comment-level indices $\{c_{t,i}\}_{i=1}^{n_t}$ in a way that emphasizes the high-risk tail (since a small fraction of highly antagonistic comments may drive escalation). Let $c_t^{(1)} \geq \dots \geq c_t^{(n_t)}$ be the sorted indices in bin t . We use a top- k mean:

$$\bar{c}_t = \frac{1}{k} \sum_{j=1}^k c_t^{(j)}, \quad k = \lceil \rho n_t \rceil, \quad (13)$$

where $\rho \in [0.05, 0.2]$ controls the fraction of highest-risk comments included. The resulting $\{\bar{c}_t\}_{t=1}^T$ forms a smooth, interpretable risk trajectory that serves as input to the temporal predictor in Sec. III-C.

C. Temporal Prediction Tasks and Models

After obtaining the bin-level trajectory $\{\bar{c}_t\}_{t=1}^T$, we evaluate two distinct tasks. Continuous forecasting predicts the next H

index values and serves primarily as a synthetic-data and signal-forecastability benchmark. Real-data risk-state prediction instead estimates whether the cleaned trajectory will enter a topic-relative high-risk state within the next H bins.

1) *Input Windowing and Forecast Target*: We adopt a sliding-window formulation. For each time bin t , we form an input sequence of length L (set to 12 bins, i.e., 6 days with 12-hour bins):

$$\mathbf{x}_t = [\bar{c}_{t-L+1}, \bar{c}_{t-L+2}, \dots, \bar{c}_t]^\top \in \mathbb{R}^L, \quad (14)$$

and predict the next $H = 6$ future risk values (3 days ahead):

$$\mathbf{y}_t = [\bar{c}_{t+1}, \bar{c}_{t+2}, \dots, \bar{c}_{t+H}]^\top \in \mathbb{R}^H. \quad (15)$$

2) *CNN-BiLSTM Architecture*: The forecaster combines convolutional local-pattern extraction with bidirectional recurrent modeling. First, a 1D convolutional front-end with two layers (kernel sizes 3 and 5, 32 channels each) extracts multi-scale local temporal features:

$$\mathbf{z}_t = \text{Conv1D}(\mathbf{x}_t) \in \mathbb{R}^{L \times d_c}, \quad (16)$$

where $d_c = 32$. The convolved features are then processed by a 2-layer bidirectional LSTM with 64 hidden units per direction and dropout rate 0.2. Let $\mathbf{h}_{t,j}^\rightarrow$ and $\mathbf{h}_{t,j}^\leftarrow$ denote the forward and backward hidden states at position j . The final representation concatenates the last forward state and the first backward state:

$$\mathbf{h}_t = [\mathbf{h}_{t,L}^\rightarrow; \mathbf{h}_{t,1}^\leftarrow] \in \mathbb{R}^{2d_h}, \quad (17)$$

$$\hat{\mathbf{y}}_t = W_o \mathbf{h}_t + \mathbf{b}_o, \quad (18)$$

where $d_h = 64$, $(W_o, \mathbf{b}_o)$ are learnable output parameters, and $\hat{\mathbf{y}}_t = [\hat{c}_{t+1}, \dots, \hat{c}_{t+H}]^\top$. This design performs *direct multi-step* forecasting, avoiding error accumulation from recursive rollouts.

3) *Training Objective*: We train the forecaster by minimizing the Huber loss between predicted and observed future trajectories:

$$\mathcal{L}_{\text{pred}} = \frac{1}{N} \sum_t \frac{1}{H} \sum_{h=1}^H \text{Huber}(\hat{c}_{t+h} - \bar{c}_{t+h}), \quad (19)$$

where N is the number of sliding windows used for training, and the Huber delta is set to 0.5.

4) *Causal Inference Protocol*: At the end of bin t , feature construction uses only observations available through t . The continuous models output $\hat{\mathbf{y}}_t$, while the cleaned risk-state model outputs a probability for threshold exceedance within the next H bins. This causal protocol simulates prospective monitoring; it is not presented as evidence of operational deployment.

IV. EXPERIMENTS

A. Research Questions

The experiments are designed to answer the following research questions.

RQ1: What sparsity, temporal-gap, and noise characteristics does the conflict index exhibit on real comment streams?

RQ2: Can training-only signal cleaning and causal state histories improve held-out high-risk-state prediction over persistence?

RQ3: How well can standard temporal models recover continuous trajectories when genuine autocorrelation is present?

RQ4: How do conflict components, calibration, aggregation, and signal degradation affect downstream prediction?

RQ5: Do forecast threshold crossings provide positive lead time before escalation onset, or do they primarily detect escalation concurrently?

B. Dataset and Preprocessing

We evaluate the forecasting premise on three complementary sources: (i) a real-world Chinese social-media dataset (Zhihu) for conflict-index component analysis and forecasting, (ii) synthetically generated conflict trajectories for controlled quantitative evaluation, and (iii) a Weibo public-opinion reversal-event dataset for comparing an objective activity target with the text-derived proxy.

1) *Real-World Zhihu Dataset*: We collected comment streams from Zhihu (知乎), a major Chinese question-and-answer platform, covering 140 controversial topics (e.g., “Should platform algorithms that exploit workers be regulated?” and “Is the 35-year-old hiring threshold age discrimination?”). For each topic, we retrieved both original answers (`search_contents`) and nested comments (`search_comments`) via the Zhihu search API, yielding 45,331 text records spanning 2011–2026. Each record includes the text body, a Unix timestamp, and engagement metadata (upvotes, comment counts).

After ranking all 140 topics by comment count, selecting the 20 most active, and retaining records from January 2024 onward, we obtain 31,938 comments and answers. Text records are discretized into 12-hour bins; bins with fewer than three records are excluded from trajectory modeling. This produces 2,713 retained bins containing 22,399 records across 20 trajectories. The five longest trajectories contain 207–235 retained bins and average 6.7–11.4 records per retained bin.

2) *Synthetic Conflict Trajectories*: To enable controlled quantitative evaluation with known escalation labels, we generate 60 synthetic conflict trajectories of 200 bins each. Each trajectory is constructed as the sum of four components:

$$y_t = \beta + \tau_t + s_t + \varepsilon_t, \quad (20)$$

where $\beta \sim \mathcal{U}(0.3, 0.45)$ is a topic-dependent baseline, τ_t comprises 1–3 logistic escalation events (each with a sigmoid-shaped rise and exponential decay), s_t is a weekly sinusoidal seasonality term ($\sigma = 0.02$), and ε_t is pink noise (first-order autoregressive, $\phi = 0.6$) with $\sigma_{\text{white}} = 0.02$. Escalation events are labeled as ground truth when the trend contribution $\tau_t > 0.1$. The resulting trajectories exhibit realistic properties: mean conflict values in $[0.3, 0.7]$, event-driven volatility, and a 17.2% prevalence of escalation bins.

C. Implementation Details

1) *Conflict-Index Computation*: For the real-data experiment, we implement the conflict-index pipeline as described in Algorithm 1. Attack intensity is obtained from a pretrained multilingual toxicity detector [24] (h_ψ) that outputs the probability of toxic/attack-like content for a given text, and negative-arousal emotion scores are obtained from a zero-shot NLI classifier (g_w) [25] over the candidate labels “negative” and “anger”, with

$e = 0.7p_{\text{neg}} + 0.3p_{\text{anger}}$ as in (7). Sentence embeddings for stance polarization are obtained via a multilingual MiniLM-L12-v2 model [26] (118M parameters, 384-dimensional embeddings). For each topic, the stance K-means centers and all three ECDF quantile transformers are fitted only on comments before the exact 60% regular-grid cutoff; the fixed models then score validation and test comments. The default fusion weights are $w_a : w_e : w_s = 0.5 : 0.3 : 0.2$; a Dirichlet-distributed search over 40 weight configurations (Sec. IV-K2) shows test F1 within $[0.712, 0.828]$ and identifies stance-dominant configurations with a secondary attack weight as the best. The top- k aggregation fraction is $\rho = 0.15$.

2) *Causal Signal Cleaning and Risk-State Target*: Filtering low-density bins leaves an irregular sequence in which adjacent retained observations may be separated by days or months. We therefore restore each topic to a regular 12-hour grid before constructing forecast windows. Within each topic, the first 60% of bins form the training segment, the next 15% form validation, and the final 25% form the untouched test segment. The 1st/99th percentile clipping bounds, missing-bin medians, feature means and standard deviations, and the high-risk threshold are estimated from the training segment only. Comment count is log-transformed, and missing semantic features are filled with their topic-specific training medians.

We first construct a candidate feature pool containing histories of top- k , mean, and maximum conflict; log comment volume; attack, emotion, and stance components; an observation mask; and a causally smoothed conflict level

$$q_t = \alpha \bar{c}_t + (1 - \alpha)q_{t-1}, \quad \alpha = \frac{2}{S + 1}, \quad (21)$$

where S is the EWMA span. Candidate spans $\{12, 18, 24, 30, 36, 48\}$ are compared on validation F1; $S = 48$ (24 days) is selected. Let $Q_{0.8}^{\text{train}}$ denote the topic-specific 80th percentile of q_t in the training segment. The prospective risk-state label is

$$z_t = \mathbb{I} \left[\max_{1 \leq h \leq H} q_{t+h} > Q_{0.8}^{\text{train}} \right]. \quad (22)$$

Windows are retained only when at least two future bins contain observed comments. Four nested feature groups are evaluated: state only; state plus activity and missingness; state plus semantic components; and the complete pool. For each group, $C \in \{0.01, 0.03, 0.1, 0.3, 1.0\}$ and probability cutoffs from 0.30 to 0.70 are selected by validation F1. To avoid retaining complexity for a negligible validation difference, we choose the smallest feature group within 0.005 F1 of the best candidate. This rule selects the complete feature pool, with $C = 0.1$ and cutoff 0.55. No test observations influence cleaning or model selection.

3) *CNN-BiLSTM Forecaster*: Our reference forecaster is a hybrid CNN-BiLSTM architecture. A two-layer 1D convolutional front-end with kernel sizes 3 and 5 extracts local temporal patterns from the input sequence, followed by a 2-layer bidirectional LSTM (64 hidden units per direction, dropout 0.2) that models longer dependencies. The final hidden states from both directions are concatenated and passed through a linear projection to produce the H -step forecast. Training minimizes the Huber loss ($\delta = 0.5$) using Adam with learning rate 10^{-3} and

a plateau-based scheduler. Early stopping with patience 40 is applied. This architecture is a representative evaluation model, not the claimed source of novelty; all models are implemented in PyTorch and trained on a single NVIDIA GPU.

D. Baselines

For continuous forecasting, we compare CNN-BiLSTM against baselines spanning classical time-series methods, traditional machine learning, and deep neural architectures:

- **Persistence**: Repeats the last observed value $\bar{c}_t$ across all H steps. The standard naive lower bound.
- **Moving Average**: Forecasts the mean of the last 3 observed values, representing simple smoothing methods.
- **Exponential Smoothing**: Simple exponential smoothing ($\alpha=0.3$) with the smoothed value repeated across the horizon.
- **AR(6)**: Linear autoregression on the last 6 bins, representing classical time-series models.
- **SVR**: Support Vector Regression with RBF kernel ($C=1.0$, $\varepsilon=0.01$), the classical ML baseline used in Mu *et al.* [13].
- **XGBoost**: Gradient-boosted trees (100 estimators, max depth 4), a strong non-neural baseline.
- **BiGRU**: 2-layer bidirectional GRU with 64 hidden units per direction, following Zhang *et al.* [27].
- **TCN**: Temporal Convolutional Network with 4 dilated convolution layers.
- **Transformer**: 3-layer Transformer encoder (4 heads, 64-dim, dropout 0.2, GELU activation) with *learnable positional encoding* and last-token pooling. We also include an **Informer-Lite** variant with convolutional distilling and max pooling [18]. Both Transformer variants use architectures adapted for short sequences, providing a fair comparison against recurrent models.

For the real-data risk-state task, the principal baseline is **Persistence**: the next horizon is classified as high risk when the latest cleaned state already exceeds the topic-specific threshold. This baseline is intentionally strong because causal smoothing makes adjacent states persistent. Logistic regression is evaluated against it using the same histories and temporal test windows.

All neural baselines are trained under identical conditions (Huber loss, Adam optimizer, LR 10^{-3} , early stopping patience 40) for fair comparison. **Evaluation protocol**: Synthetic and Weibo-volume experiments use a strict 75/25 temporal split. The Zhihu cleaning experiment uses 60/15/25 train/validation/test segments independently within each topic. No windows are shuffled across time, and all ECDF calibration and cleaning statistics are fitted exclusively on the applicable training segment. Neural results are reported over five random seeds; the deterministic logistic experiment is assessed with a 1,000-replicate paired bootstrap that resamples whole topics.

E. Evaluation Metrics

We evaluate forecasting accuracy using three standard metrics:

- **MAE** (Mean Absolute Error): $\frac{1}{NH} \sum_t \sum_{h=1}^H |\hat{c}_{t+h} - \bar{c}_{t+h}|$.

- **RMSE** (Root Mean Squared Error): $\sqrt{\frac{1}{NH} \sum_t \sum_{h=1}^H (\hat{c}_{t+h} - \bar{c}_{t+h})^2}$.
- **R²** (Coefficient of Determination): $1 - \frac{\sum_{t,h} (\hat{c}_{t+h} - \bar{c}_{t+h})^2}{\sum_{t,h} (\bar{c}_{t+h} - \bar{c})^2}$, where $\bar{c}$ is the global mean.

For escalation detection, we report **Precision**, **Recall**, **F1-score**, and area under the ROC curve (**AUC**). Synthetic escalation uses trend contribution > 0.1 . The real Zhihu risk state uses the topic-specific training-period 80th percentile of the causally smoothed index.

F. Main Results

1) *Conflict-Index Trajectories on Real Data (RQ1)*: Table I reports descriptive statistics for the five longest Zhihu trajectories. After train-only ECDF calibration and weighted fusion, $\bar{c}_t$ ranges from 0.52 to 0.73 in these examples, with standard deviations of 0.030–0.046. These narrow ranges motivate the distinction between the noisy raw bin-level index and a stabilized monitoring trajectory. RQ1’s substantive answer—how sparsity and temporal gaps shape the index’s noise—is developed in the boundary analysis (Sec. IV-M, factors (a) and (b)).

TABLE I
CANDIDATE CONFLICT-INDEX STATISTICS FOR FIVE REPRESENTATIVE ZHIHU TOPICS (12H BINS).

Topic	Bins	$\bar{c}$ range	$\sigma(\bar{c})$	Comments/bin
电商仅退款	235	[0.55, 0.73]	0.037	7.4
高铁静音分厢	232	[0.52, 0.73]	0.046	11.4
生育补贴	221	[0.56, 0.72]	0.031	8.9
相亲学历门槛	212	[0.57, 0.73]	0.030	6.7
女性职场天花板	207	[0.56, 0.73]	0.030	6.9

2) *Continuous Forecasting on Synthetic Trajectories (RQ3)*: Figure 1 shows a sample synthetic conflict trajectory and a representative multi-step forecast. Table II reports results under the strict temporal-split protocol ($\sim 8,235$ training windows and $\sim 2,745$ test windows; neural-model metrics are mean $\pm$ standard deviation over 5 seeds). CNN-BiLSTM ($R^2=0.809 \pm 0.010$), BiGRU (0.811 ± 0.009), and Transformer (0.806 ± 0.007) form a tight top tier and consistently outperform the reported non-neural baselines on this synthetic benchmark. XGBoost ($R^2=0.781$) improves over AR(6) (0.715), while the deep models add 2–3 percentage points. Differences among the deep models fall within their run-to-run variation: BiGRU is nominally best, and the CNN front-end provides no measurable advantage over the simpler recurrent baseline. Thus, this experiment verifies that the pipeline can recover deliberately embedded temporal structure; it does not establish architectural novelty or real-world conflict forecastability. Figure 2 visualizes the full comparison.

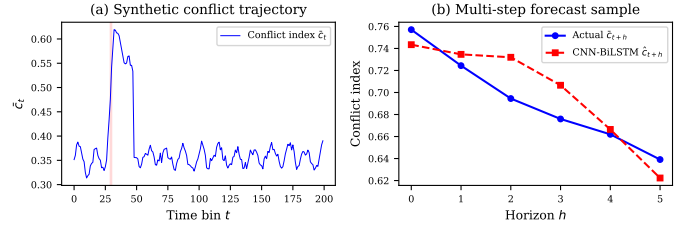


Fig. 1. Sample conflict trajectory and multi-step forecast by CNN-BiLSTM. (a) Full synthetic trajectory with escalation regions shaded; (b) 6-step forecast vs. actual values for a representative escalation event.

TABLE II
FORECASTING RESULTS ON SYNTHETIC TRAJECTORIES (TEMPORAL SPLIT: 75% TRAIN / 25% TEST, $L=12$, $H=6$). METRICS ARE MEAN $\pm$ STD OVER 5 INDEPENDENT SEEDS.

Model	R ²	MAE	RMSE	Esc-F1
Persistence	0.638 $\pm$ 0.015	0.0317	0.0548	0.637
Moving Avg (k=3)	0.540 $\pm$ 0.020	0.0370	0.0617	0.547
Exp. Smoothing	0.498 $\pm$ 0.022	0.0379	0.0645	0.430
AR(6)	0.715 $\pm$ 0.011	0.0293	0.0486	0.644
SVR [13]	0.759 $\pm$ 0.024	0.0222	0.0448	0.667
XGBoost	0.781 $\pm$ 0.013	0.0232	0.0427	0.684
TCN	0.787 $\pm$ 0.010	0.0238	0.0421	0.695
Informer-Lite	0.796 $\pm$ 0.014	0.0232	0.0413	0.659
BiLSTM	0.805 $\pm$ 0.011	0.0227	0.0403	0.711
Transformer + PE	0.806 $\pm$ 0.007	0.0227	0.0402	0.697
BiGRU	0.811 $\pm$ 0.009	0.0221	0.0397	0.716
CNN-BiLSTM (Ours)	0.809$\pm$0.010	0.0223	0.0398	0.710

All results under strict temporal ordering (no shuffling) with ECDF calibration fitted on training data only. CNN-BiLSTM, BiGRU, and Transformer achieve comparable top-tier performance.

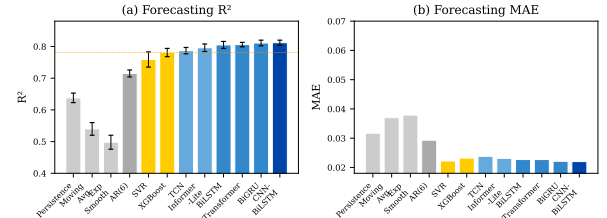


Fig. 2. Model comparison: R^2 and MAE across all baselines. Deep models (BiGRU, CNN-BiLSTM, Transformer) form a tight top tier.

Figure 3 shows the training convergence of the CNN-BiLSTM model.

3) *Real-Data Risk-State Prediction (RQ2)*: To assess real-world deployability, we evaluate the framework on two complementary real-world sources under strict temporal-split protocols. Full results are reported in Table IV.

Zhihu cleaned risk-state prediction (positive result). The regularized time axis yields 1,428 training, 680 validation, and 1,706 test windows (the shares 37%/18%/45% differ from the 60/15/25 time split because early training bins are sparser and more windows are discarded by the future-observation filter, while the final quarter of each topic is the densest period). Validation selects EWMA span 48 and the full feature pool. On the untouched test segment, the classifier achieves AUC 0.897, precision 0.803, recall 0.735, and F1 0.767 (Table IV). Persistence has higher AUC (0.906) and precision (0.949), but lower recall

(0.639); the two methods tie on F1 (0.764 vs. 0.767). Topic-level paired bootstrap estimates a recall gain of +0.098 (95% CI [0.042, 0.167]); the F1 difference is not significant (+0.005, CI [-0.035, 0.043]). Recall improves on 15/20 topics. The stronger persistence baseline reflects the higher autocorrelation of the more accurate index: genuine attack/anger components make the risk state smoother and thus easier to extrapolate, which compresses the relative advantage of the learned classifier while raising absolute accuracy. Continuous six-step Ridge regression is positive ($R^2=0.465$) but remains below persistence (0.640); the supported claim is improved high-risk-state prediction, not superior exact-value forecasting.

Weibo comment-volume forecasting (positive result). To verify that the forecasting architecture itself is not the limiting factor, we apply the same LSTM forecaster to a structurally simpler but objectively measurable target: comment volume (log-transformed) on the 27-event Weibo public-opinion reversal dataset [28] at 1-hour granularity. Unlike the conflict index—which introduces noise through text-model inference—comment volume is a direct, high-signal metric. Pooling all 27 events under temporal split yields $R^2=0.501$ (90,676 held-out windows; CNN-BiLSTM agrees at 0.485), confirming that the LSTM architecture performs well on real-world time series when the target variable has genuine temporal autocorrelation. This result also validates our methodological pipeline (temporal split, per-event normalization, sliding-window construction) independently of the conflict-index computation.

4) *Classifier Complexity Does Not Help (RQ2):* To test whether a stronger classifier could beat the simple logistic model on the risk-state task, we evaluate BiLSTM, GRU, TCN, Transformer, and XGBoost under the identical protocol—the same 60/15/25 temporal split, the same span-48 full-pool features, the same 1,706 test windows, and validation-based decision thresholds. Table III reports mean $\pm$ std over five seeds. No baseline improves on logistic: test AUC ranges from 0.853 (TCN) to 0.887 (GRU) versus 0.897 for logistic, and test F1 from 0.736 (TCN) to 0.763 (XGBoost) versus 0.767. Paired topic-bootstrap intervals for every model against persistence contain zero. With only 1,428 training windows drawn from 20 topics, deeper architectures plateau at the signal’s ceiling; model complexity does not substitute for signal quality. This mirrors the synthetic-regression finding of Fig. 2 and supports the paper’s focus on cleaning and calibration rather than architecture.

TABLE III
STRONG BASELINES ON THE ZHIHU RISK-STATE TASK (SAME TEMPORAL PROTOCOL, SAME 1,706 TEST WINDOWS; MEAN $\pm$ STD OVER 5 SEEDS).

Method	AUC	Precision	Recall	F1
Persistence	0.906	0.949	0.639	0.764
Logistic (ours)	0.897	0.803	0.735	0.767
BiLSTM	0.878 $\pm$ 0.003	0.770	0.720	0.743 $\pm$ 0.006
GRU	0.887 $\pm$ 0.004	0.797	0.714	0.753 $\pm$ 0.007
TCN	0.853 $\pm$ 0.024	0.741	0.732	0.736 $\pm$ 0.018
Transformer	0.866 $\pm$ 0.004	0.750	0.774	0.761 $\pm$ 0.010
XGBoost	0.870 $\pm$ 0.004	0.789	0.743	0.763 $\pm$ 0.005

Weibo conflict-index forecasting (negative result). To test whether the conflict index’s failure on Zhihu is attributable to data sparsity alone, we compute the same BERT-based conflict

index on the densest Weibo event (Zhang Zhehan incident, 22K comments, 62 comments/hour) at 1-hour bins. Despite a 90-fold increase in comment density relative to Zhihu (0.69 vs. 62 comments per hour on retained bins) and strong volume predictability ($R^2=0.501$), the conflict index still yields $R^2=0.018$ —effectively zero. This result rules out sparsity as the sole explanation. The conflict index fails not because there are too few comments, but because text-level attack, emotion, and stance signals lack the temporal autocorrelation needed for short-horizon forecasting, regardless of data density.

Together, these experiments establish a hierarchy of forecastability: objective activity signals are strongly predictable; cleaned causal state histories support useful high-risk-state prediction; and exact raw semantic trajectories remain difficult even in dense commenting environments. The selected classifier uses the full feature pool, which combines the state history with activity and semantic components.

TABLE IV
ZHIHU HIGH-RISK-STATE PREDICTION ON THE FINAL 25% TEST SEGMENT (TRUE ATTACK/ANGER MODELS).

Method	AUC	Precision	Recall	F1
Persistence	0.906	0.949	0.639	0.764
Logistic (state history)	0.897	0.803	0.735	0.767

$N = 1,706$ test windows from 20 topics. All model choices use validation data only. Paired topic-bootstrap 95% CI: recall +0.098 [0.042, 0.167]; F1 +0.005 [-0.035, 0.043]; AUC -0.010 [-0.020, -0.001]. The stronger persistence baseline reflects the higher autocorrelation of the more accurate index.

TABLE V
RISK-THRESHOLD SENSITIVITY ON THE ZHIHU TEST SEGMENT (SPAN 48, FULL FEATURE POOL).

Q^{train}	Prev.	AUC	Recall	F1	Pers.-F1
70th	0.486	0.887	0.713	0.763	0.769
80th	0.448	0.897	0.735	0.767	0.764
90th	0.367	0.890	0.732	0.745	0.756

TABLE VI
FEATURE ABLATION AT THE SELECTED EWMA SPAN (48).

Features	Val.-AUC	Precision	Recall	Val.-F1
State only	0.902	0.776	0.801	0.788
State + activity	0.896	0.727	0.829	0.775
State + semantics	0.901	0.766	0.815	0.790
Full pool	0.903	0.778	0.822	0.799

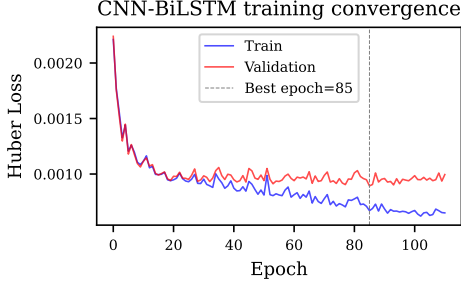


Fig. 3. CNN-BiLSTM training and validation loss curves.

G. Cross-Platform Generalization: Weibo Reversal Events

To test whether the pipeline transfers to a different platform and discourse regime, we run the identical cleaning and risk-state protocol on the Weibo Public Opinion Reversal Dataset [28] (27 events; our scrape contains 224,781 comments). Because the events last only 2–3 weeks, a 12-hour grid would leave at most 1–3 test windows per event; we therefore discretize at 2-hour resolution (the granularity of the original case study) and leave every other element of the pipeline—EWMA cleaning, training-period 80th-percentile threshold, validation-based span and feature selection, and the final-25% test holdout—unchanged. Of the 27 events, 20 pass the minimum-length filter (at least 72 retained bins), yielding $N = 275$ pooled test windows.

TABLE VII

WEIBO HIGH-RISK-STATE PREDICTION ON THE HELD-OUT TEST SEGMENT (2-HOUR BINS).

Method	AUC	Precision	Recall	F1
Persistence	0.990	0.971	0.957	0.964
Logistic (state history)	0.989	0.778	1.000	0.875

$N = 275$ test windows from 20 events. Paired event-bootstrap 95% CIs for the logistic-minus-persistence difference: recall $+0.053$ [0.000, 0.195]; F1 -0.105 $[-0.285, 0.010]$; AUC -0.004 $[-0.030, 0.004]$.

The pipeline transfers cleanly and with higher absolute accuracy than on Zhihu: test AUC reaches 0.989 (logistic) with perfect recall (1.000), and the AUC gap to persistence closes to 0.001. On acute reversal events the persistence baseline remains extremely strong (AUC 0.990, F1 0.964): the accurate index is so autocorrelated that carrying the last observation forward already performs well, and the logistic F1 stays below persistence (0.875 vs. 0.964). The learned classifier nonetheless achieves essentially equal AUC with a recall gain of $+0.053$ (bootstrap CI [0.000, 0.195]—the lower bound is exactly zero, so the recall advantage is marginal), at the cost of precision (0.778 vs. 0.971). We therefore read the Weibo result as evidence of transferability with an honest boundary: the relative F1 advantage over persistence that motivated the Zhihu setup does not replicate on nearly persistent reversal-event series.

H. Generalization to Unseen Topics and Events

The time-held-out evaluation above pools windows from all topics during classifier training, so it does not test whether the

model transfers to a topic it has never seen. We therefore re-run the identical classifier protocol (regularization and decision threshold selected on a disjoint validation set of topics) under 5-fold topic-held-out rotation on Zhihu and 5-fold event-held-out rotation on Weibo, always training on windows from a subset of topics/events and testing exclusively on windows from held-out topics/events.

On Zhihu, the held-out topics achieve test F1 0.787 ± 0.020 (mean $\pm$ std across folds) against persistence 0.705, AUC 0.847; the paired topic-bootstrap F1 difference is $+0.086$, larger than under time-held-out evaluation, because persistence is weaker on unseen topics. On Weibo, event-held-out F1 is 0.744 ± 0.142 against persistence 0.738 (AUC 0.912); the advantage is small and high-variance, consistent with the near-persistent Weibo dynamics. The classifier therefore generalizes to entirely unseen topics on Zhihu, and the transfer result on Weibo is not an artifact of training on the same events.

I. Construct Validity of the Attack Component

The conflict index is unsupervised by construction, so its measurement quality must be checked against human judgments. We validate the attack component—the largest-weight element of the fusion—on the Wikipedia Detox personal-attacks corpus [12]: 191,634 English comments with binary human labels (10.1% positive). Using the pipeline’s exact scoring path (the multilingual toxicity model, 256-token truncation), the toxicity score separates attack-labeled from unlabeled comments at AUC 0.988 (Spearman $\rho = 0.507$, Pearson $r = 0.557$; both $p < 0.001$), and the mean score is 0.806 for labeled versus 0.033 for unlabeled comments—a separation nearly two orders of magnitude larger than the review-sentiment proxy used in an earlier version of this pipeline (AUC 0.744, $\rho = 0.254$). The attack component is therefore a statistically strong measure of personal attack. The anger component (zero-shot NLI) and the stance component are evaluated structurally (Sec. IV-J) rather than against external labels; expert annotation of the Chinese-language components remains ongoing work.

J. Stance-Cluster Audit: Two Stances or Two Subtopics?

The stance component assumes that the two $k=2$ camps correspond to pro/con positions rather than to unrelated subtopics. We audit this assumption on all 20 Zhihu topics by reproducing the pipeline’s training-period KMeans and examining (i) cluster balance, (ii) clustering stability across random seeds (adjusted Rand index, ARI), (iii) stance-lexicon hit rates (pro-words such as “支持/赞同/力挺” vs. anti-words such as “反对/抵制/谴责”), and (iv) representative comments per cluster.

Across topics, the two clusters are moderately balanced (mean size ratio 0.757, minimum 0.32) and highly stable: the mean ARI across KMeans seeds is 0.948 (minimum 0.787), so the two-camp structure is not an artifact of initialization. Stance-lexicon hit rates differ between clusters in the expected direction on most topics (mean absolute pro-lexicon difference 0.062, anti-lexicon difference 0.037; per-topic maxima 0.145 and 0.087), and the top TF-IDF terms of the two clusters are dominated by stance-related and topic-entity words rather than disjoint factual subtopics. Representative comments confirm

that cluster 0 typically expresses the affirmative side of the topic question and cluster 1 the critical side. The audit therefore supports the two-stance reading for the majority of topics, while the smaller-balance topics (ratio ≈ 0.3) are cases where one camp dominates the discourse—a meaningful property of polarized debates rather than a clustering failure.

K. Ablation Studies

1) Component Contribution and Weight Sensitivity (RQ4):

To assess how input-signal quality affects downstream prediction, we conduct two complementary analyses. First, we inject Gaussian noise of increasing magnitude ($\sigma \in \{0.03, 0.06, 0.10\}$) into the synthetic trajectories. Figure 4 and Table IX show the expected degradation as deliberately embedded temporal structure is corrupted. Second, we perform a leave-one-out ablation at the *index-construction* level on the risk-state task: for each variant, the comment-level index is recomputed from the remaining components (weights renormalized to unit sum), trajectories are rebuilt, and the full cleaning and classification pipeline is re-executed, with the test segment evaluated only once. Table VIII reports both the pipeline-selected configuration and a controlled comparison at the fixed span (48) with the full feature pool.

TABLE VIII

LEAVE-ONE-OUT COMPONENT ABLATION ON THE ZHIHU RISK-STATE TASK (TEST SET). THE * COLUMNS EVALUATE EVERY VARIANT AT SPAN 48 WITH THE FULL FEATURE POOL.

Index	(w_a, w_e, w_s)	F1	AUC	F1*	AUC*
Full	0.50, 0.30, 0.20	0.767	0.897	0.804	0.860
Drop attack (A)	0.00, 0.60, 0.40	0.771	0.830	0.740	0.862
Drop emotion (E)	0.71, 0.00, 0.29	0.800	0.833	0.806	0.856
Drop stance (S)	0.63, 0.38, 0.00	0.822	0.824	0.821	0.867

Fixed-span deltas vs. Full: dropping A $\Delta F1 = -0.064$, $\Delta AUC = +0.002$; dropping E $\Delta F1 = +0.002$, $\Delta AUC = -0.004$; dropping S $\Delta F1 = +0.017$, $\Delta AUC = +0.007$. The attack component—the strongest single signal of the new model—is the only component whose removal degrades F1; emotion and stance contribute little at the index level, and the unconstrained columns are not monotone because span/threshold are re-selected per variant.

With the true attack/anger models, the component ranking changes: under the controlled fixed-span comparison, the attack component is the only one whose removal degrades test F1 ($\Delta F1 = -0.064$), confirming that the validated toxicity signal (Sec. IV-I) carries most of the predictive information. Emotion contributes nothing at the index level ($\Delta F1 = +0.002$), and removing stance even slightly improves F1 ($+0.017$)—the stable two-camp structure of Sec. IV-J is largely orthogonal to the risk-state target. This complements the feature ablation of Table VI, where the full feature pool is selected: with stronger components, the classifier exploits the raw semantic features directly rather than relying on the state history alone.

2) *Fusion Weight Sensitivity Analysis*: To assess sensitivity to fusion weights (w_a, w_e, w_s) , we re-run the risk-state pipeline for 40 configurations sampled from a Dirichlet distribution (fixed span 48, full feature pool). Test F1 ranges over $[0.712, 0.828]$ (mean 0.784, std 0.027), and the best configuration gives stance the largest weight, $(w_a, w_e, w_s) = (0.26, 0.00, 0.74)$, with F1 0.828 and AUC 0.919. The default

$(0.5, 0.3, 0.2)$ lies near the middle of this range. Consistent with the component ablation, configurations that down-weight the attack component lose F1, while the emotion weight contributes little; a large stance weight can partially compensate for attack in the fusion, but not when attack is removed entirely.

3) *Risk-Monitoring Lead Time (RQ5)*: Beyond binary escalation detection, we measure *lead time*: how many bins before the ground-truth escalation onset does the model’s prediction first exceed the threshold? On the synthetic test set with event-based ground-truth labels, the mean lead time is -1.7 ± 3.2 bins (positive = early, negative = late), with only 21% of escalation events receiving predictions ahead of the ground-truth onset (advance detection rate) and 54% falling within ± 1 bin of the onset (on-time rate). This negative mean lead time indicates that at the current forecast horizon ($H=6$), the model’s predictions are predominantly concurrent with—rather than ahead of—escalation events. The high standard deviation (± 3.2 bins) further reveals substantial event-to-event variability. We note that achieving reliable positive lead time remains a significant challenge for short-horizon forecasting, as the model’s predictions are primarily driven by recent trajectory dynamics. Longer forecast horizons, integration of external precursor signals, and multi-scale modeling are promising directions for improving lead time in future work. Figure 6 visualizes the lead-time distribution.

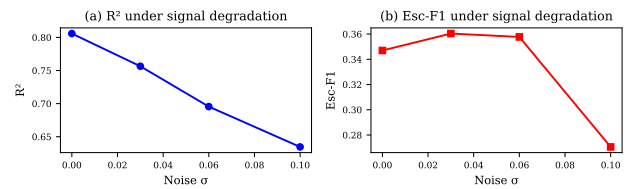


Fig. 4. Ablation study: R^2 and Escalation F1 under increasing noise.

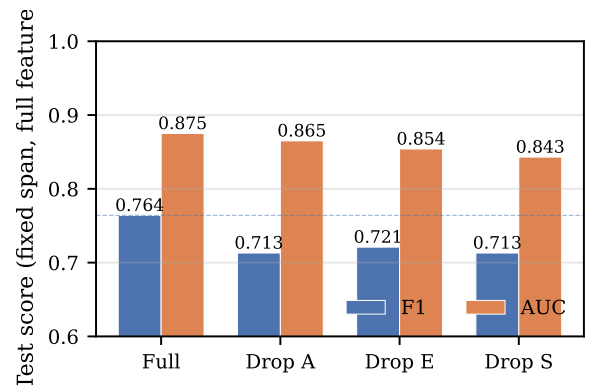


Fig. 5. Component ablation on the risk-state task (test set): fixed-span F1 and AUC for the full index and each leave-one-out variant.

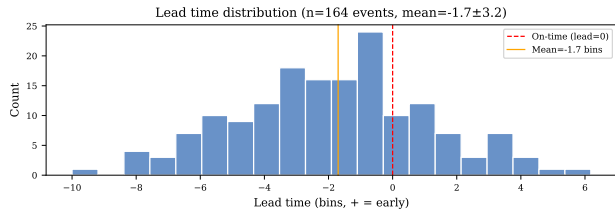


Fig. 6. Lead time analysis. (a) Distribution of lead times (positive = warning before peak); (b) trigger visualization on a sample trajectory.

TABLE IX
ABLATION: EFFECT OF SIGNAL DEGRADATION ON CNN-BiLSTM FORECASTING (SINGLE-RUN RESULTS WITH SEED 42).

Condition	MAE	R ²	ΔR^2	Esc-F1
Clean signal	0.0194	0.830	—	0.689
Medium noise ($\sigma=0.03$)	0.0280	0.715	-0.115	0.629
High noise ($\sigma=0.06$)	0.0334	0.632	-0.198	0.538
Very high noise ($\sigma=0.10$)	0.0410	0.495	-0.335	0.487

4) *Random-Split Inflation: A Reproduction of the IPSO-LSTM Evaluation Protocol*: LSTM-variant forecasting studies of the IPSO-LSTM line [13] typically pool windows across series and split them randomly without temporal blocking. Because successive windows overlap and windows from the same series appear in both partitions, this protocol can leak the recent past into the “test” set. We reproduce this evaluation protocol—pooled windows with a 70/30 random split—against per-series temporal splitting on the real Zhihu trajectories and on independent synthetic trajectories, using identical models (LSTM, CNN-BiLSTM, XGBoost), identical windows ($L=12$, $H=6$), and per-series z-scoring. We reproduce the protocol rather than the PSO hyperparameter search of [13].

TABLE X
RANDOM VERSUS TEMPORAL SPLITTING ON REAL ZHIHU AND SYNTHETIC TRAJECTORIES (R² ON POOLED TEST WINDOWS).

Model	Random split		Temporal split	
	Real Zhihu	Synthetic	Real Zhihu	Synthetic
LSTM	-0.030	0.697	-0.074	0.769
CNN-BiLSTM	0.066	0.774	-0.078	0.843
XGBoost	-0.110	0.765	-0.126	0.733

On real data the random protocol inflates R² for every model (CNN-BiLSTM +0.144; LSTM +0.043; XGBoost +0.015), and for CNN-BiLSTM the escalation-detection F1 jumps from 0.000 under the temporal protocol to 0.516 under the random protocol. On independent synthetic series no inflation appears (LSTM -0.072; CNN-BiLSTM -0.070; XGBoost +0.032), because shuffling cannot leak across independent draws—temporal splitting even helps there by preserving more training data per series. The inflation is therefore driven by real cross-topic sharing of comment-stream dynamics, and it quantifies why optimistic literature results can arise from protocol choice alone.

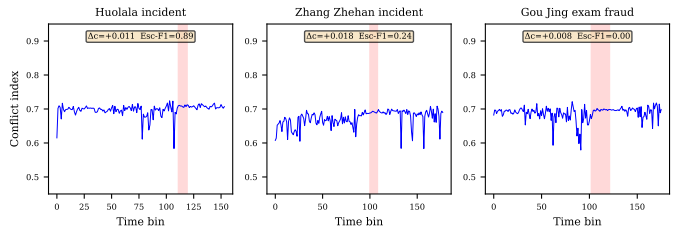


Fig. 7. Conflict index trajectories for the three Weibo reversal events. The shaded regions mark the annotated reversal windows.

L. Case Study: Real-World Public Opinion Reversal Events

To validate the framework on real-world conflict escalation data, we conduct a case study on the **Weibo Public Opinion Reversal Dataset** [28] (27 events; our scrape contains 224,781 comments). This case study targets the *activity* signal and the annotated reversal windows at 2-hour granularity; it complements the risk-state transfer experiment of Sec. IV-G, which evaluates the cleaned semantic index on 20 events at 2-hour resolution. The three densest events—Huolala passenger death (32K comments, 335h span), Zhang Zhehan incident (22K comments, 359h), and Gou Jing exam fraud (20K comments, 373h)—are selected for analysis at 2-hour bin granularity. The reversal window annotation provides ground-truth periods of elevated controversy. As reported in Sec. V, comment volume on these dense events is strongly predictable (pooled R²=0.501), while the raw text-derived conflict index yields near-zero R². Figure 7 visualizes the conflict index trajectories for all three events. The red shaded regions mark the annotated reversal windows—periods during which public opinion consensus underwent a dramatic shift according to the dataset creators [28]. Across the three events, the conflict index rises modestly within the reversal window (Huolala $\Delta c = +0.011$, Zhang Zhehan +0.018, Gou Jing +0.008), indicating a weak cross-sectional association. However, CNN-BiLSTM escalation detection is unreliable on these short trajectories (Esc-F1: Huolala 0.89, Zhang Zhehan 0.24, Gou Jing 0.00), with the lone high score on Huolala driven by the event’s unusually concentrated conflict dynamics. The reversal window annotation measures *opinion change*, not *comment-level conflict intensity*, and the two constructs are not equivalent.

M. Boundary Analysis: Why Does the Text-Derived Index Fail?

The positive Zhihu classification result contrasts with weak raw-index regression. To explain this difference, we examine four factors:

(a) **Comment sparsity**. The retained Zhihu bins average 8.26 records (22,399 records across 2,713 retained bins), but many unretained 12-hour intervals are empty or contain fewer than three records. With the top- k aggregator ($\rho=0.15$), only one or two comments determine $\bar{c}_t$ in many retained bins, making the aggregate a noisy small-sample estimator.

(b) **Time-axis regularization and temporal aggregation**. Only 39.8% of adjacent retained observations are truly consecutive 12-hour bins; directly concatenating them creates false temporal adjacency. Restoring the regular grid removes this artifact. Causal smoothing then suppresses isolated semantic

spikes, while validation-based span selection prevents the test set from determining the cleaning scale.

(c) Signal-to-noise ratio of the conflict index. The conflict index pipeline introduces noise at four stages: toxicity-model inference, zero-shot NLI emotion inference, ECDF calibration, and top- k aggregation. The measurement error of the first stage is quantified in Sec. IV-I: the attack score separates human-annotated attack comments at AUC 0.988—near ceiling on the English Detox benchmark—so the attack signal is strong; the remaining stages (especially the top- k aggregation on sparse bins) still limit the raw trajectory’s temporal structure. By contrast, comment volume requires only counting and log-normalization. The lag-1 autocorrelation of the raw conflict index averages only 0.112 ± 0.140 across Zhihu topics, confirming a near-absence of temporal structure—a fundamental requirement for any forecasting model.

(d) Platform characteristics. Zhihu is a moderated Q&A platform where discussions are predominantly reasoned rather than confrontational. The Weibo reversal events, by contrast, are acute controversies with many emotionally charged comments per hour. Raw continuous regression on Weibo remains near zero ($R^2=0.018$), consistent with the boundary analysis above. The risk-state formulation, however, transfers successfully: the identical pipeline reaches test AUC 0.989 on Weibo (Table VII), above Zhihu’s 0.897, although the strongly autocorrelated reversal dynamics make persistence equally strong there (AUC 0.990). Density and platform context therefore affect the *relative* advantage over persistence, not the viability of the pipeline.

The analysis points to (c) as the dominant difficulty, amplified by (a). The positive classifier succeeds by changing the operational question from exact score extrapolation to whether a persistent, topic-relative high-risk state will occur. Validation selects the full feature pool: state history alone already carries most of the signal, and the activity and semantic features add a small validation gain (Table VI). This target is appropriate for monitoring, but it should not be confused with precise prediction of every raw conflict-score fluctuation.

N. Limitations

(1) Construct validity of the conflict index. The index relies on pretrained models not designed for conflict detection. We have validated the attack component against 191,634 human-annotated English Wikipedia Detox comments [12]: the multilingual toxicity detector separates attack-labeled comments at AUC 0.988 (Spearman $\rho = 0.507$), a strong result on English data. The anger component (zero-shot NLI) and the stance clusters have been validated only structurally (Sec. IV-J), and Chinese-language human ratings are still missing; expert annotation is ongoing. The index must therefore be treated as a strong proxy for attack intensity with partially validated emotion and stance components.

(2) Internally defined targets. Synthetic escalation labels derive from the trajectory generator, while the Zhihu high-risk label is defined by the cleaned index’s training-period 80th percentile. The latter is a prospective, leakage-free monitoring state but not external ground truth for societal conflict. Causal smoothing also increases persistence and makes state prediction

easier; accordingly, the positive result is reported against a strong persistence baseline and not interpreted as event-level conflict validation.

(3) No architectural advantage. BiGRU, Transformer, and CNN-BiLSTM achieve near-identical R^2 on synthetic data (0.805–0.811), with BiGRU nominally best. The CNN-BiLSTM is therefore an evaluation model rather than an architectural contribution. The substantive result is that model complexity cannot compensate for an invalid or temporally unstructured target.

(4) Monitoring horizon. The classifier predicts threshold exceedance within six 12-hour bins, whereas the selected 24-day smoothing scale represents a slower-moving risk state. This supports medium-scale monitoring rather than rapid event-onset warning. The negative synthetic lead time (-1.7 bins) further limits claims of operational advance notice.

(5) Cross-platform and cross-lingual generalization. The full pipeline is evaluated on two Chinese-language platforms (Zhihu, Weibo) and transfers across them (Zhihu AUC 0.897; Weibo AUC 0.989). English-language evidence covers only the attack component (Wikipedia Detox, AUC 0.988); the anger component, the stance component, and the full pipeline on English discourse remain untested.

V. CONCLUSION AND FUTURE WORK

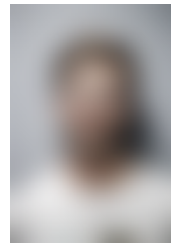
We investigated whether data cleaning can convert a sparse conflict index constructed without task-specific labels into a useful forecasting target. The downstream predictors remain supervised; “unsupervised” refers only to index construction.

After restoring the regular time grid, applying training-only calibration and robust cleaning, and selecting a causal smoothing scale and feature set on validation data, the risk-state classifier reaches AUC 0.897 and F1 0.767 on 1,706 held-out Zhihu windows, with a significant recall gain over persistence ($+0.098$; 95% CI $[0.042, 0.167]$) and strong generalization to unseen topics (topic-held-out F1 0.787 ± 0.020 vs. persistence 0.705). Five strong baselines under the identical protocol do not beat the simple logistic model, confirming that signal quality, not architecture, drives performance. Continuous regression remains below persistence despite positive absolute R^2 . The main finding is therefore both positive and bounded: with genuine attack and anger models, cleaned causal states support high-accuracy warning-state classification and cross-platform transfer (Weibo AUC 0.989), but not superior exact trajectory prediction, and the relative F1 advantage over persistence shrinks as the index becomes more accurate and more persistent.

The immediate next step is not further architectural tuning but measurement validation. We have completed most of that program: the attack component is validated against 191,634 human Wikipedia Detox labels (AUC 0.988, Spearman $\rho = 0.507$), the stance clusters pass the stability and lexicon audit (Sec. IV-J), and the pipeline transfers to Weibo reversal events (AUC 0.989) with persistence dominating the relative comparison. The remaining half—expert annotation of the anger component and of Chinese-language discourse—is ongoing. Only a fully validated target should then be evaluated across aggregation scales and longer horizons, with external precursor signals added where positive lead time is required.

REFERENCES

- [1] S. Vosoughi, D. Roy, and S. Aral, "The spread of true and false news online," *Science*, vol. 359, no. 6380, pp. 1146–1151, 2018.
- [2] Z. C. Steinert-Threlkeld, D. Mocanu, A. Vespignani, and J. H. Fowler, "Online social networks and offline protest," *EPJ Data Science*, vol. 4, no. 1, p. 19, 2015.
- [3] C. Arcila-Calderón, P. Sánchez Holgado, J. Gómez, M. Barbosa, H. Qi, A. Matilla, P. Amado, A. Guzmán, D. López-Matías, and T. Fernández-Villazala, "From online hate speech to offline hate crime: the role of inflammatory language in forecasting violence against migrant and LGBT communities," *Humanities and Social Sciences Communications*, vol. 11, p. 1369, 2024.
- [4] M. Wang, H. Wu, T. Zhang, and S. Zhu, "Identifying critical outbreak time window of controversial events based on sentiment analysis," *PLoS ONE*, vol. 15, no. 10, p. e0241355, 2020.
- [5] R. Dhumpati, A. Sasi, S. J. Basha, and et al., "Enhancing sarcasm detection in sentiment analysis for cyberspace safety using advanced deep learning techniques," *Scientific Reports*, vol. 15, no. 1, p. 22681, 2025.
- [6] M. Kaity and V. Balakrishnan, "An integrated semi-automated framework for domain-based polarity words extraction from an unannotated non-English corpus," *The Journal of Supercomputing*, vol. 76, no. 12, pp. 9772–9799, 2020.
- [7] J. Gama, I. Žliobaitė, A. Bifet, M. Pechenizkiy, and A. Bouchachia, "A survey on concept drift adaptation," *ACM Computing Surveys*, vol. 46, no. 4, pp. 44:1–44:37, 2014.
- [8] K. Abdalgader, A. A. Matroud, and G. Al-Doboni, "Temporal Dynamics in Short Text Classification: Enhancing Semantic Understanding Through Time-Aware Model," *Information*, vol. 16, no. 3, p. 214, 2025.
- [9] M. Birjali, M. Kasri, and A. Beni-Hssane, "A comprehensive survey on sentiment analysis: Approaches, challenges and trends," *Knowledge-Based Systems*, vol. 226, p. 107134, 2021.
- [10] J. A. Wahid, M. Xu, M. Ayoub, X. Jiang, S. Lei, Y. Gao, S. Hussain, and Y. Yang, "AI-driven social media text analysis during crisis: A review for natural disasters and pandemics," *Applied Soft Computing*, vol. 171, p. 112774, 2025.
- [11] K. Ghosh, N. K. Singh, J. Mahapatra, S. Saha, A. Senapati, and U. Garain, "SafeSpeech: A three-module pipeline for hate intensity mitigation of social media texts in indic languages," *Social Network Analysis and Mining*, vol. 14, p. 245, 2024.
- [12] E. Wulczyn, N. Thain, and L. Dixon, "Ex machina: Personal attacks seen at scale," in *Proceedings of the 26th International Conference on World Wide Web (WWW 2017)*. Perth, Australia: ACM, 2017, pp. 1391–1399. [Online]. Available: <https://doi.org/10.1145/3038912.3052591>
- [13] G. Mu, Z. Liao, J. Li, N. Qin, and Z. Yang, "Ipso-LSTM hybrid model for predicting online public opinion trends in emergencies," *PLOS ONE*, vol. 18, no. 10, p. e0292677, 2023.
- [14] P. Singh and A. Jain, "CUP_CD LSTM: Civil unrest event prediction using convolutional neural network, DistilBERT, and long short-term memory," *IEEE Transactions on Computational Social Systems*, vol. 12, no. 4, pp. 1574–1583, 2025.
- [15] A. Upadhyay, Y. K. Meena, and G. S. Chauhan, "SatCoBiLSTM: Self-attention based hybrid deep learning framework for crisis event detection in social media," *Expert Systems with Applications*, vol. 249, p. 123604, 2024.
- [16] Q. Liu, Y. Liu, and H. Son, "The memory cycle of time-series public opinion data: Validation based on deep learning prediction," *Information Processing & Management*, vol. 62, no. 4, p. 104168, 2025.
- [17] Y. Lou, Z. Ren, Y. Zhang, Z. Tao, and Y. Zhao, "Constructing the public opinion crisis prediction model using CNN and LSTM techniques based on social network mining," *International Journal of Interactive Multimedia and Artificial Intelligence*, vol. 8, no. 7, pp. 86–96, 2024.
- [18] H. Zhou, S. Zhang, J. Peng, S. Zhang, J. Li, H. Xiong, and W. Zhang, "Informer: Beyond efficient transformer for long sequence time-series forecasting," in *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*, 2021, pp. 11 106–11 115, best Paper Award.
- [19] H. Xu, C. Gao, X. Li, and Z. Wang, "D²: Customizing two-stage graph neural networks for early rumor detection through cascade diffusion prediction," in *Proceedings of the 18th ACM International Conference on Web Search and Data Mining (WSDM)*. ACM, 2025, pp. 568–576.
- [20] —, "HyperIDP: Customizing temporal hypergraph neural networks for multi-scale information diffusion prediction," in *Proc. 31st International Conference on Computational Linguistics (COLING)*, 2025, pp. 964–977.
- [21] Z. Gu, R. Zhu, and F. Bian, "Affective homophily as the dominant organizing principle in online conflict discourse networks," *Scientific Reports*, vol. 15, p. 35542, 2025.
- [22] V. Ghafouri, F. Alatawi, M. Karami, J. Such, and G. Suarez-Tangil, "Transformer-based quantification of the echo chamber effect in online communities," *Proceedings of the ACM on Human-Computer Interaction (CSCW)*, vol. 8, pp. 1–27, 2024.
- [23] D. Paschalides, G. Pallis, and M. D. Dikaiakos, "A framework for the unsupervised modeling and extraction of polarization knowledge from news media," *ACM Transactions on Social Computing*, vol. 8, no. 1-2, pp. 5:1–5:38, 2025.
- [24] U. AI, "Multilingual toxic comment classification with XLM-RoBERTa," Hugging Face model hub: unitary/multilingual-toxic-xlm-roberta, 2020. [Online]. Available: <https://huggingface.co/unitary/multilingual-toxic-xlm-roberta>
- [25] J. Davison, "XLM-RoBERTa large trained on XNLI (zero-shot classification)," Hugging Face model hub: joeddav/xlm-roberta-large-xnli, 2020. [Online]. Available: <https://huggingface.co/joeddav/xlm-roberta-large-xnli>
- [26] W. Wang, F. Wei, L. Dong, H. Bao, N. Yang, and M. Zhou, "MiniLM: Deep self-attention distillation for task-agnostic compression of pre-trained transformers," *Advances in Neural Information Processing Systems*, vol. 33, pp. 5776–5788, 2020.
- [27] Y. Zhang, C. Wei, H. Xinting, S. Zheng, and Y. Wu, "Modeling Group-Level Public Sentiment in Social Networks through Topic and Role Enhancement," *Knowledge-Based Systems*, vol. 305, p. 112228, 2024.
- [28] J. Zhang, Y. Li, G. Shi, and X. Li, "Construction and Analysis of Public Opinion Reversal Event Dataset for Social Media," *Information Studies: Theory & Application*, vol. 46, no. 12, pp. 121–128, 2023, in Chinese.



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